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Contested Spatialities of Digital Sovereignty

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ABSTRACT

"Digital sovereignty" has become a buzzword in digital policies. Contrary to the imaginary of digital transformation as preceding an era of limitless global networking in the 1990s, approaches to state regulation and delimitation of data flows as well as programmes for national digital infrastructures are justified with calls for digital sovereignty across very different contexts. This forum brings together contributions from political geography, law, computer science, and ethics that compare and analyse discourses and practices of digital sovereignty. The case studies on Russia and the EU reveal parallels as well as fundamental differences in the conception and implementation of digital sovereignty. Essays on the challenges posed by new forms of cross-border interaction (such as cloud computing) and new actors (such as digital platforms) illustrate that the traditional coupling of concepts of sovereignty, territoriality and the state, of jurisdiction and borders, must be rethought. The essays in this forum thus make it clear that the digital transformation is not simply a socio-technical modernisation process. It is rather shaped in specific ways and should be understood and analysed as (geo)-political discourses and practices. The forum contributes to the development of a *political digital geography* that analyses how the digital transformation is contested and produced in specific ways and unearths the politics and spatialities conceived and produced in these discourses and practices.

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Introduction: The Emerging Geopolitics of Data

Georg Glasze, Frédérick Douzet, Amaël Cattaruzza, Finn Dammann

The production of digital data has accelerated tremendously over recent decades. More and more processes are being datafied. At the same time, these data is the basis for more and more applications. Until a few years ago, the growth of data flows and the broader digital transformation were often described as accelerators of a global integration and the formation of a post-territorial world. There have been predictions that flows and networks will replace the traditional territorial order of states (emblematic of this perspective is Friedman 2007, while geographers early on criticised these predictions as too crude, e.g., Tuathail 1999).

Since the early 2010s, however, this vision of an open and networked world has become increasingly cracked (see also Bratton 2015: XVIII). The ability to collect, control and exploit data has appeared as a form of leverage for political, economic, social and strategic power. Between 2007 the present day, cyberattacks, ranging from the Estonia denial-of-service attack in May 2007 to the rise of ransomware such as WannaCry and NotPetya in 2017, have multiplied. In addition, states, as well as private actors (GAFAM and BATX, among others) have developed their digital capabilities, or *cyberpower* (Nye 2010), to act on the international scene (espionage, attack, destabilisation, surveillance) in a more or less secret and visible way (Buchanan 2020; Dunn Cavelty 2012). Increasingly, and in very different contexts, the free flow of data has come to be perceived as a threat. As a result, political and academic debates on data flows have become increasingly securitised, territorialised and ultimately geopoliticised (see also Lambach 2019). Multiple national initiatives have emerged to regain control over strategic data, such as policies of data localisation or reshaping of the architecture of connectivity. This fundamental shift has lead to increasing demands for "digital sovereignty" from different governments and different people. With calls for a stronger role for the state, for strategic autonomy and digital borders in political contexts as different as China, Russia, India and the EU (see the contributions in this forum as well as e.g. Budnitsky and Jia 2018; Vila Seoane 2021; Zeng, Stevens, and Chen 2017) the discourses and practices of a digital sovereignty can be seen as a condensation and materialisation of these new geopolitics of data flows.

Studies in geography dealt early on with how digital technologies change sociospatial relationships (e.g. Graham 2000; Sui and Morrill 2004) and thus may alter the ways in which societies are governed. However, surprisingly few studies in geography have focused on digitisation itself as a produced and thus contested sociotechnical process (for exceptions see Warf 2012; Zook and Graham 2007), let alone a representation of and tool in geopolitical rivalries

(Cattaruzza 2019; Douzet 2020). In this context, the question of how the digital transformation is politically produced and discursively embedded, and which spaces are thereby newly articulated, seems to us to be central to the debates, demands, and strategies of a "digital sovereignty" in various geographical contexts.

To open up these complex questions to a broader debate in geography we are bringing together contributions from two interdisciplinary Franco-German workshops in 2020 and 2021 (for further contributions on the digital sovereignty discourse from a media science perspective see e.g. Couture and Topin 2019, and for political sciences perspectives see e.g. Pohle and Thiel 2020).

The Quest for Digital Sovereignty

In the early 2000s, authoritarian governments began to propagate concepts of "digital sovereignty" often with reference to ideas of a Westphalian world order of territorially defined sovereign states (see Zeng, Stevens and Chen 2017; Budnitsky and Jia 2018; Limonier 2018 and the contribution of Pétiniaud et al. on Russia). Since the 2010s, narratives of "digital sovereignty" have been taken up also in many other political spheres – not least in Europe (see the contribution of Glasze et al.). The current configuration of the digital transformation is seemingly being increasingly problematised as a threat to "the sovereign state" – and in some contexts also explicitly to "the sovereign subject" (see the contribution of Winkler and Dammann).

From a political-geographical perspective, the discourses of "digital sovereignty" and the practices legitimised by them can be read as elements of geopolitical visions that problematise specific threats, articulate specific images of an enemy, and, not least, (re)produce spatialities. The buzzword "digital sovereignty" is demonstrably often used to legitimise decidedly territorial closure strategies for a future shaping of data flows – e.g. in Russia but also to some extent in other parts of the world (see for example Möllers 2020). At the same time, these closure strategies are inevitably in tension with processes of global interconnectedness. It is therefore not surprising that the buzzword "digital sovereignty" is also used to legitimise strategies for enforcing technological and/or regulatory standards with a potentially global impact as specifically geoeconomic strategies – for example, in the European Commission's priorities for the period 2019–2024 (see Braun & Hummel's contribution on EU AI policies).

Now, authors such as Krasner (2001) and Agnew (2005) pointed out a few years ago that "sovereignty" should be analysed not as an objective sociospatial structure but rather as a political concept or even a myth – albeit a concept that guides and legitimises political practice. The notion of "digital sovereignty" is in this sense the perfect illustration of why "sovereignty" should be analysed as

a geopolitical representation carried out by different actors, that depends on the context and can serve a specific political objective (see Desforges and Danet 2020 and the following contributions). Although the search for *one* meaning of sovereignty is therefore not very productive, an examination of the genealogies of the concept can help us better understand the paradoxes and challenges of the contemporary "digital sovereignty" debates.

Sovereignty is a modern concept that has sometimes been interpreted as a transfer or "secularisation" of the Christian idea of an omnipotent God to worldly, i.e. social, conditions (Bartelson 1995; Schmitt 2008 [2008 [1922]). Modern state theorists have developped a conception of sovereignty as a supreme, non-derived, all-encompassing, externally and internally unrestricted rule over a specific territory, peculiar to the state and its representatives. The modern international order of states is still based on this territorial concept. Therefore, all forms of networking across territories risk coming to a state of tension with territorial concepts of sovereignty (Sassen 1996). Cross-border data flows are ultimately a challenge to a traditional territorial understanding of sovereignty. At the same time, the intensive networking in the digital age has made it clear that sovereignty, as the ability to act, is ultimately always dependent on relational embedding. A way of thinking that ties sovereignty to physically delimited and isolated entities inevitably falls short here (see also Bratton 2015; Pistor 2020).

With the Enlightenment, the idea of unrestricted sovereignty became contained. Concepts of popular sovereignty were directed against the idea of unrestricted princely rule – the sovereignty should be with "the people". The concept of popular sovereignty presupposes the ideal of rational, informed, autonomous, self-determined and in that sense "sovereign" subjects. In liberal and plural democracies, legislative power is limited by guarantees of individual rights, principles of constitutionalism and a separation of powers. Therefore, ideas of sovereignty as the unrestricted rule of the state are inevitably in tension with ideas of open, plural and democratic societies. This tension explains the different representations of the concept of "sovereignty" in authoritarian and more democratic-liberal contexts – which are reproduced within different claims for digital sovereignty (on the tension between "state sovereignty" and "popular sovereignty", see Benhabib 2016, with reference to the debate on digital sovereignty see Thiel 2019).

For legal studies, but also for political sciences, sovereignty is primarily associated with the idea of a legal claim to autonomous rule from within and non-interference from outside. However, it has often been pointed out that the absoluteness of this claim is a normative ideal. Against this backdrop, works from the historical sciences and political geography have argued that sovereignty should also be thought of as a capacity produced in and through specific sociomaterial practices. Thus, various studies have shown, for example, that the modern knowledge instruments of statistics and modern cartography were

co-constitutive with the formation of centralised and territorial state-sovereign rule (Branch 2014; Elden 2013; Hannah 2000). Against this background, the current debate on digital sovereignty can in some ways be read as an expression of the tension between legal and political claims and socio-material practices in the context of sociotechnical relations in the digital transformation.

Outlook: Perspectives on the Contested Spatialities of Digital Sovereignty

The following contributions highlight the spatialities articulated in discourses and practices of digital sovereignty: Bertran, Limonier, and Pétiniaud, for example, show that the government of Russia has made "digital sovereignty" in the 2010s a leitmotif of digital policy but has encountered numerous difficulties in its implementation (e.g. territorial strategies for the development of a "national Internet" and the development of a domestic digital industry). Somewhat surprisingly, an examination of the discourses and practices of digital sovereignty in France, Germany, and the EU (see the contribution by Glasze et al.) reveals some parallels (almost reflexive demands for territorial closures after the Snowden revelations, promotion of a European digital industry). On the other hand, numerous voices in Europe are trying to merge the geopolitical demands for digital sovereignty with the notion of "European values" and in doing so are also invoking the concept of an individual digital sovereignty of its citizens (see the contribution by Winkler and Dammann). More recently, the European Commission seems to be increasingly linking the buzzword of digital sovereignty with a geoeconomic strategy that designs European regulatory approaches with a preferably global impact. Such a European approach is legitimised, for example, in the design of artificial intelligence applications with reference to the ultimately often vague "European values" (see the contribution by Braun and Hummel). Demands for digital sovereignty, however, do not inscribe themselves exclusively as competition between states. As Grumbach and Zanin show, it is precisely the power of private digital platforms to control society that is increasingly being problematised as a challenge to the power of states to exercise their sovereign powers and control society. In this context, geopolitical confrontations intersect with geoeconomic competition in a complex way. The analysis of platforms as geopolitical objects illustrates that concepts and practices of sovereignty, boundaries and territories may have to be fundamentally rethought in the digital age. Finally, the contribution by Nicolai and Géry sheds light on the challenge for the territorially organised legal system. Using the example of criminal law, they discuss how the legal system reacts to the challenge of transnational data flows: To what extent can demands for digital sovereignty

be understood here, on the one hand, as demands by states for transnational access to data and, on the other hand, as the protection of individuals' personal data?

The multiple problematisations of sovereignty in the digital sphere seem to us to be paradigmatic of the current geopoliticisations of data: Therefore,our forum also wants to contribute to the development of a *political digital geography* that analyses how the digital transformation is contested and produced in specific ways and questions the politics and spatialities conceived and produced in these processes – within an interdisciplinary dialogue (like in this forum with contributions from geography, law, computer sciences and ethics).

Russia's Pursuit of "Digital Sovereignty": Political, Industrial and Foreign Policy Implications and Limits

Louis Pétiniaud, Kevin Limonier, Marie-Gabrielle Bertran

Introduction

By the early 2010s, following a tendency observed in authoritarian states (see Introduction), digital sovereignty began to emerge as an important element of the Russian public debate. In March 2012, Vladimir Putin inaugurated his third term with a "conservative turn", reaffirming Russia as the protector of "moral values" and presenting the Internet as a threat to his political project and the stability of the regime. The memory of the role played by social networks in the genesis and coordination of the Arab Spring in 2010–2011 was still vivid. In November 2012, Russia denounced U.S. hegemony over the Internet and suggested that sovereign states represented in the General Assembly should gain control over its governance. This proposal contravened the multistakeholder governance model on which the Internet has been based since its foundation and was curtly rejected by the United States and its allies. A few months later, in June 2013, Edward Snowden revealed to the world a gigantic network surveillance programme initiated by the NSA. This thunderclap gave important visibility to Russia's denunciation of American digital hegemony and fuelled narratives that presented the Internet as a vector of influence and destabilisation to the benefit of the West.

Between the denunciation of the U.S. hegemon, the defence of moral values and the imperatives of national security, the Internet became the object of a triple convergence in Russia, leading to intense legislative production that was still in progress at the beginning of the 2020s. In the name of "digital sovereignty", several laws have provided the state with increasingly powerful means of controlling the network. In 2014, the Russian State Duma (the lower house) adopted the first legislation on data localisation, which requires operators and

platforms to host the data of Russian individuals or legal entities in the territory of the Russian Federation. Two years later, the so-called Yarovaya laws made it mandatory for platforms to store users' metadata for three years and to set up backdoors for intelligence services in addition to communicating their decryption keys to those services. Finally, a new level was reached in 2019 with the "sovereign RuNet" law, which gives the authorities technical and legal means to control data packets entering and leaving Russian territory and to "isolate" the Russian network from the rest of the world when necessary.

Although it is almost impossible to grasp all the effects of this 10-year *crescendo*, this paper intends to provide a brief overview of the implications and limits of Russia's pursuit of "digital sovereignty", focusing on three major issues. First is the compliance of the Russian IT sector, which is crucial for embedding the software and tools needed to guarantee the country's independence in critical fields. Second is the topographical and topological structure of the Russian Internet, which is one of the most complex networks in the world. Third is the effects of Russia's pursuit of digital sovereignty in the post-Soviet area.

The Russian IT Sector: A Major but Not Always Compliant Asset

RuNet is characterised by a rich ecosystem of digital applications that are commercialised by a few key players in the Russian economy. These players, among which we can distinguish Yandex and Mail.Ru, or the Rambler group, are now concentrating the main capitalisations in the Russian digital market. They are considered to be systemic economic actors by the Russian authorities, who have managed to arrogate the right to oversee their board of directors and internal functioning.

The compliance of the private digital sector has indeed become indispensable for the application of the new digital policies adopted by the Russian government in the past few years, following the principle of "digital sovereignty". These policies notably imply a *russification* of digital solutions in the country. However, the importance of the private sector in enforcing them poses serious limitations to the power of the state. Moreover, the externalisation of state digital capacities induces a decentralisation of Russia's digital infrastructures and a proliferation of relevant stakeholders for their management. This phenomenon increases the number of attack vectors through unsecured access points and puts at risk the data that are stored or travel through these infrastructures. The cyberattack disclosed in July 2019 against a Russian company called SyTech, which was under contract with the FSB, is revealing of these limits. Some companies also use the argument of cybersovereignty merely as a marketing tool to sell their products to public institutions (Gerasimov 2020).

Finally, some private actors simply refuse to or cannot comply with sudden changes in the legislation. The 2014 law on data storage relocation was met with strong reluctance by many Russian and foreign companies,

which formerly stored their users' data abroad for several reasons (costs, technical convenience, etc.). Their reluctance led to the adoption of several addenda to the law to allow all the players one more year to comply. Additionally, a number of actors, such as small Internet service providers (ISPs), use different strategies to avoid complying with specific regulations, such as in the domain of mass surveillance (Ermoshina and Musiani 2017).

The Difficult Restructuring of Russia's Network

Russia's plan for consolidating the sovereignty of its network implies the restructuring of the architecture of the subnetworks that make up the Russian segment of the Internet. In other words, Russia is attempting to be able to control or at least map how data transits both within the country and towards the rest of the worldwide Internet. This kind of information is paramount for the constitution of a potential independent Russian network in case of attacks or disruption.

This challenge is herculean, considering the extraordinary complexity of Russia's technical infrastructure and subnetwork profusion. The Internet developed in Russia during the 1990s in a very anarchic way, without strong state supervision. As a result, the nationwide network is the third-largest and most complex in the world, with exactly 6,555 subnetworks (owned by different actors such as Internet service providers, data centres, public institutions such as universities and hospitals, and private companies). In comparison, Germany and France each have approximately 2,000 subnetworks. This technical complexity is also visible at the ISP level: Russia has thousands, whereas France has about a dozen.

While Russia's high level of Internet entropy has contributed to its resilience, it has made control of the network all the more difficult. For instance, to enforce the "sovereign RuNet" law, RKN relies on the willingness of Russia's thousands of autonomous systems and ISP administrators to cooperate. However, some are unwilling to do so, as there is a systemic lack of trust in the authorities (Ermoshina, Loveluck, and Musiani 2021). In this context, creating a potentially isolated RuNet is a tremendous task that requires great efforts of both a technical and a political nature.

Russia's Digital Sovereignty as a Foreign Policy Asset in the near Abroad?

The Russian pursuit of digital sovereignty has notable implications for the post-Soviet space. Following the break-up of the Soviet Union, the postimperial nature of the former USSR made post-Soviet countries more or less reliant on Russia, including regarding cyberspace. As the successor state of the Soviet Union, the Russian Federation has inherited economic, industrial, educational and cultural prevalence as well as significant infrastructural centrality in the region.

Furthermore, Russia, a country with a "post-imperial syndrome" (De Tinguy 2009), has been eager since the 2000s to maintain or reinforce its influence in the post-Soviet space, which Russian leaders do not view as entirely "foreign" (Mankoff 2009) and sometimes refer to as their "near abroad" (Tuathail 2016).

Consequently, in the field of cyberspace, in terms of infrastructure and applications but also in other domains,[1] Russia remains an influential hub and a centripetal force in the post-Soviet space, albeit with notable exceptions.[2] The power and leverage that Russia holds over post-Soviet countries through cyberspace has tended to increase with Russia's legislative inflation. Intermediation platforms, on the one hand, and Internet infrastructure and data routing, on the other, are two domains where the implications of the Russian drive for digital sovereignty are crucial for the post-Soviet space.

Intermediation platforms are Internet applications (websites and apps) that ensure intermediation between consumers (users) and producers of services or goods, such as Facebook, Amazon or Google, that lead the market (see the contribution "Platforms vs. States: A Sovereignty Conundrum"). In the post-Soviet space, Russian language platforms created in the 1990s, such as Yandex, Mail.ru and VKontakte, have been adopted in most of the Russian-speaking post-Soviet space (Limonier 2018), where the Russian language remains widely used. This predominance has given Russia enhanced capabilities in terms of digital surveillance and commercial influence. The share of Russian platforms is, however, declining, sometimes for political reasons. In post-Soviet countries closer to the West or antagonistic to Russia (the Baltic states, Ukraine, Moldova, Georgia), Russian platforms are losing ground to their Western counterparts or to platforms from other countries (*e.g.* China).

In terms of infrastructure, the Russian legislation that reinforces political control over Internet routes has increased Russia's leverage in terms of digital surveillance but also created weaknesses for post-Soviet countries in terms of connectivity resilience, making potential Internet disruptions easier. Russia is indeed the central node of post-Soviet digital networks, due mostly to centralised transport and energy infrastructures along which most Internet infrastructures are built. This centrality increases Russia's potential control over safe and secure Internet access in neighbouring countries. Furthermore, post-Soviet disputed territories, such as Crimea, Donbass, or Abkhazia, that constitute geopolitical leverage by Moscow have become connectivity enclaves that are connected to the Internet solely through Russia, giving Moscow the ability to easily control or disrupt the local Internet, including in the eventuality of a kinetic conflict (Pétiniaud and Salamatian 2019). In some cases, the structure of Internet routes has become an integrated element of geopolitical rivalries that shape the post-Soviet space.

Conversely, dynamics of infrastructural multilateralism can be observed. Regions such as Central Asia rely mostly on north-south Internet backbones that connect them to the rest of the world through Russian infrastructures. In

this context, significant efforts are being made by countries such as Georgia, Azerbaijan and Kazakhstan to make their networks less dependent on Russia. Some of these initiatives rely on China's digital Silk Road initiative (Kozłowski 2018), as this project would ultimately create an east-west backbone that would mitigate Russia's influence (Castillo 2021).

Conclusion

Russia's pursuit of a sovereign Internet is an ambitious project with no precedent at this scale. Countries with networks tailored to this ambition, such as China and Iran, have profited from practical preconditions: China has developed its network with this specific end in mind, while for political and structural reasons, it has been much simpler to rebuild the Iranian network in this way. Russia, in contrast, has not only a very dense and complex network but also a scattered yet strong civil society and a private sector that are not always eager to comply with regulations.

Reception and Elaboration of "Digital Sovereignty" in Three European Discourse Arenas: France, Germany, and the EU

Georg Glasze, Finn Dammann, Max Münßinger, Didier Danet, Clotilde Bômont, Alix Desforges

Introduction: the Vision of a "Global Information Society" – and Critiques

As the volume of digital data streams grew by leaps and bounds in the 1990s, the idea of a free flow of digital information and the vision of a "global information society" gained traction in many political contexts in Europe (Glasze and Dammann 2021). A report developed in 1994 under the then EU commissioner for the internal market and industrial affairs, Martin Bangemann, linked ideas of an integration of Europe into such a "global information society" with promises of economic prosperity, democratisation and social progress. The report called for policies of deregulations and privatisations of previously state-organised telecommunications in EU member states (see also Mueller 2017; Reiberg 2018; Schaper-Rinkel 2006).

However, since the late 2000s, an increasing number of voices have questioned the vision of a borderless "global information society" – also within the EU. The ideas of a free circulation of data, regulated only to a very limited extent by the state, are losing plausibility in the context of numerous debates about data protection, IT security, cybercrime, copyright, digital mass surveillance, big tech oligopolies and national competitiveness. Under buzzwords

such as "digital sovereignty", "technological sovereignty" and "data sovereignty", calls are now being made for stronger state regulation of data flows and more national autonomy in the digital sphere.

In this contribution, we outline how the buzzword digital sovereignty has been taken up and processed in three different but linked discursive spheres in Europe: policy oriented debates at the EU level and public debates on a national level in the two countries that seem to have most intensively shaped "European" visions of "digital sovereignty" thus far – France and Germany.

The Emerging Quest for "Digital Sovereignty" in Germany and France

The first mentions of "digital sovereignty" in France (*souveraineté numérique*) date back to the mid-2000s but remain little known to the general public. At the time, the historical role of the US in Internet governance was increasingly being criticised. In this context, Bernard Benhamou, a member of the French delegation to the World Summit on the Information Society (2003–2005), and the political advisor Laurent Sorbier published an article in which they developed the idea of an European "middle position", in between a pro-status quo (the "multi-stakeholder model" that ultimately secures a prominent role for the U.S.),[3] and the demands for strict State control of the Internet as advocated by Russia and China. They provided insight into the contours of a "digital sovereignty" for Europe (Benhamou and Sorbier 2006).

In the broader French public debate, "digital sovereignty" was taken up only at the beginning of the 2010s, when Pierre Bellanger, the CEO of Skyrock (a popular radio station) contributed generalising the notion with an entrepreneurial and strategic approach. Bellanger condemned the abandonment of the "French digital sovereignty" in the face of the "American imperialism" (Bellanger 2011) underlying the impact of the growing dominance of U.S. companies. He defined digital sovereignty as "the control of our present and our destiny as manifested and directed by the use of computer technologies and networks" (*ibid.*).

The Snowden revelations in 2013 were seen as the confirmation of those concerns, and the notion of "*souveraineté numérique*" became a subject of lively and intense political debate. Once again, the notion arose of condemning American hegemony over digital issues and calling for the development of a digital industry at the French and European levels. Industrial development was, indeed, seen as the means of emancipation from foreign companies (American but also, more recently, Chinese) and from the inherent geopolitical risks of growing dependency (Danet and Desforges 2020).

While in the 2010s and 2020s, media and political discourses have referred extensively to "digital sovereignty", the term does not appear as frequently in the official documents of the French government, which opt instead for the concept of "strategic autonomy" (Danet and Desforges 2020). A key concept in

French military doctrine since World War II, strategic autonomy is seen as the means of enabling the state exercise of sovereignty. This implies France's "autonomous ability to assess, decide and act". Strategic autonomy fits well with digital affairs since it does not exclude alliances – as opposed to the notion of independence that implies isolated action and is often associated with sovereignty. In fact, alliances are promoted as long as they are explicitly desired and, more importantly, when they are informed.[4]

If the first initiatives towards digital sovereignty in France were driven mainly by economic and industrial arguments, they took a security turn in the mid-2010s. Efforts to support the digital industrial sector and the emergence of technical solutions and standards were completed with normative and regulatory measures.

The French "sovereign cloud" project is symptomatic of this discursive evolution of the French debate. Cloud computing was initially perceived only as a technical object and mode of organisation of an information system. Poor assessment of the socioindustrial context and the technology itself caused the project of a "French cloud" to faile. However, over the following years, the power issues associated with cloud computing became more obvious to political actors (Bômont and Cattaruzza 2020). This shift opened the way for the French-German initiative of a "European cloud" – Gaia-X.

In the political debates in Germany, the buzzword of "digital sovereignty" was taken up on a larger scale only after the Snowden revelations of 2013 (Glasze and Dammann 2021; Misterek 2017; Steiger, Schünemann, and Dimmroth 2017). In the German context, "digital sovereignty" marks a rather fundamental shift in perspective because ideas of national autonomy and territorial separation (often associated with "sovereignty") had been rather alien to the political discourse in Germany – which to a great extent was shaped after World War II by ideas of European and global integration (Maull 2007). A closer look shows, however, that while demands for an opening towards and integration into a "global information society" prevailed in the 1990s and 2000s, other voices were already present. Against the backdrop of several public scandals about data leaks, in which personal data held by companies and public authorities proved to be insufficiently protected, progressive-liberal voices such as the Chaos Computer Club demanded in the 2000s that the state better protect its citizens in the digital space (Chaos Computer Club 2009; 2010).

However, it was only with Edward Snowden's revelations in 2013, which showed among other issues that the digital communications of top German politicians could also be monitored, that the question of whether the state can still regulate the circulation of data came into focus in public political debates in Germany. In 2014, many politicians spoke of a loss of "government capacity". In immediate reaction to the Snowden revelations, ideas of territorial closure became prominent, and concepts of a corresponding legal-technical

organisation of data flows were discussed – prominently, for example, in the idea of German routing of emails (or, alternatively, European Schengen routing; Glasze and Dammann 2021; Misterek 2017).

Even though these concepts of territorial isolation did not materialise, the buzzword "digital sovereignty" has been associated with political debates in Germany since the mid-2010s with the conviction, shared across almost the entire political spectrum that the state currently lacks the capabilities and competencies to adequately regulate data circulation and that state capacities must therefore be strengthened. In the federal government, the term has been explicitly taken up in both security policy and economic policy orientation. Large business associations in the IT and electronics industry in Germany have also adopted the buzzword "digital sovereignty": here, too, more state involvement is called for in the sense of an active industrial policy commitment to the development or safeguarding of globally competitive digital key technologies. A rather specific feature of the German-language debate is the focus on the individual level and thus on education and social policy. Thus, numerous interventions by business associations, educational institutions and, again, almost the entire political spectrum use the buzzword "digital sovereignty" precisely with regard to the idea of a digitally competent, independent and, in this sense, sovereign subject (see contribution Dammann and Winkler).

The comparison shows that the term "digital sovereignty" was used in France as early as the mid-2000s. The debate in France followed established discourses critical of U.S. hegemonies and concepts of a national industrial policy. In the German debate, "digital sovereignty" was taken up as a political concept only after the Snowden revelations. However, since 2014, a lively and multilayered debate has been developing in Germany. A common feature of the public discussions in France and Germany is the increasing invocation of a European level: since the mid-2010s, numerous voices in France and Germany have been calling for the EU to develop a European vision of "digital sovereignty".[5] In its programme for the German EU Council presidency in 2020, for example, the German government explicitly formulated the goal of establishing "digital sovereignty" as a "leitmotiv of European digital policy" (Auswärtiges Amt 2020).

Outlook: towards Strategies of a "New Digital World" under European Auspices

Until the late 2010s, the sovereignty discourse was very rarely taken up at the level of formal EU policy. Even when terms such as "digital sovereignty" or "data sovereignty" have been mentioned, the underlying policies – strengthening domestic markets, focusing on competitiveness and securing digital infrastructures – have not (yet) changed significantly compared to the context of the 1990s and 2000s (see also Pohle 2020).

This situation has been changing in the 2020s since the beginning of the new legislative period under Commission President Ursula von der Leyen. The EU is developing a digital agenda that no longer focuses solely on the single market and integration into global contexts but also resolutely aims to create European standards with a global impact. It is increasingly engaging in European digital industrial policy and digital infrastructure development and more intensively discussing strategic cybersecurity issues.[6] Using the catchphrases "technological/digital sovereignty", an explicit external agenda of European digital policy has been developed. In this agenda, the EU is deliberately bringing into play the Brussels effect described by Bradford (2020), – the ability to regulate global markets unilaterally. With digital privacy standards such as the GDPR or the cloud project Gaia-X, the EU is trying to leverage the size of its own single market and regulatory capacity to globalise its digital standards.

The commission has emphasised that Europe's "technological sovereignty" programme is by no means exclusive but is based on benefits for Europeans and their values (see also the contribution of Braun and Hummels). Any company or state that wants to play by these rules is welcomed (European Commission 2020). In the speech accompanying the publication of the document, Von der Leyen defined technological sovereignty as "the capability that Europe must have to make its own choices, based on its own values, respecting its own rules" (Von der Leyen 2020). With this framing, the European Commission attempted to define technological sovereignty as a defence of European values while concomitantly positioning itself as a champion of an open, decentralised, undivided Internet of free markets (European Commission 2021a). With this combination of de facto regulatory power and value-oriented legitimisation, the EU is building an alternative model of digital transformation, alongside those of the United States and China, as a third option in the current global geopolitical and geoeconomic battles for the profits of the digital transformation (see also Hobbs 2020; Christakis 2020).

Sovereign Power: Artificial Intelligence and Europe's Digital Sovereignty

Matthias Braun, Patrik Hummel

Introduction: governing Artificial Intelligence

Policy-makers around the world are currently seeking the best ways to deal with the opportunities and challenges of artificial intelligence (AI). In April 2021, the EU presented a governance strategy on how it intends to harness the potential of AI and prevent potential harm (European Commission 2021b). In short, the EU is taking a risk-based approach by assigning different levels of risk to different AI technologies.

The particularly notable point about this draft from the European Commission is that since work on the framework began in 2018, the Commission has consistently emphasised that it is in a race with other global policy actors. In doing so, the EU has never made a secret of the fact that developing a governance framework for AI is an integral building block to establishing the EU as a global technology leader. While the EU seems to accept that other nations are leaders in the development of new technology, it wants to be a leader in the innovative use of AI. The AI governance framework thus has had a clear geopolitical focus from the outset: by establishing the governance framework within the EU, power and influence are to be gained both within and outside the EU. The EU stated: "While this approach will unfold in the context of the global race on AI, EU policy-makers have adopted a frame of analysis to differentiate the *EU strategy* on AI from the *US strategy* (developed mostly through private-sector initiatives and self-regulation) and the *Chinese strategy* (essentially government-led and characterised by strong coordination of private and public investment into AI technologies). In its approach, the EU seeks to remain faithful to its cultural preferences and its higher standard of protection against the social risks posed by AI – in particular those affecting privacy, data protection and discrimination rules – unlike other more lax jurisdictions." (European Commission and European Political Strategy Centre, 2018).

To ensure this differentiation, the EU described its governance approach as following a clear "human-centric" (European Commission 2021a) approach to AI. At the same time, what exactly the EU means by this remains somewhat vague: "AI is not an end in itself, but a tool that has to serve people with the ultimate aim of increasing human well-being" (Communication from the Commission to the European Parliament et al., 2019). Such a human-centric approach seeks to guarantee certain moral values and legal principles via the trustworthiness of the developed systems. Thus, the European Commission stated, "The Coordinated Plan outlines the necessary policy changes and investment at Member States level to strengthen Europe's leading position in the development of human-centric, sustainable, secure, inclusive and trustworthy AI" (European Commission 2021c). Margrethe Vestager put it even more explicitly: " . . . we can pave the way to ethical technology worldwide and ensure that the EU remains competitive along the way" (European Commission 2021a).

What exactly it means to make a technology ethical remains unclear. Likewise, if we agree that human well-being must be a central goal of political action, much remains to be said about what conclusions can be drawn from this. For this reason, the idea of human-centeredness has faced a significant amount of criticism for being vague and too soft, among other reasons (Bazzano et al. 2017). Two central goals, however, are clear. First, the new governance framework for AI proclaims moral values such as human dignity

and legal principles such as human rights as the basis for how the EU deals with AI (Floridi 2021). Second, the EU not only wants to compete with other states or associations of states with certain sets of values and legal principles but also seeks to define and establish its own values and legal principles as reference points and benchmarks for the other states. The EU itself frames the new governance framework as an essential anchor of a strategy that guarantees Europe's digital sovereignty and thus simultaneously assumes global leadership over future technology development. Consequently, a massive investment in technology development *made in Europe* was announced in the fall of 2020.

Challenges of Digital Sovereignty

Digital sovereignty is not a clearly defined term; rather, many different approaches, concepts and demands are gathered under it (see the intro to this forum and Hummel et al. 2021). One central element, however, is that sovereignty requires a certain form of power in order to be established, maintained or even expanded.

That said, sovereignty is not the same as the ability to exercise arbitrary brute power. Instead, sovereignty fundamentally depends not only on features and powers of the sovereign herself but also on the addressees of her claims to sovereign power. For sovereignty to exist, it is necessary that these addressees ascribe authority to the sovereign and recognise her legitimacy. Against this background, note that the quest for digital sovereignty as the EU understands it involves the power to define rules for oneself and others within the EU, wherever else AI has an impact on European citizens (Floridi 2021), and beyond the EU by shaping technology and regulation worldwide.

While this approach might well succeed in establishing regulatory precedents and defining technical standards, it is compatible with imposing these precedents and standards on others. This in itself need not be considered unusual, but it would be fair to be explicit that such imposition could result in mere dominion over a space that cuts across technological and geopolitical landscapes and decidedly fall short of instantiating the normatively much more intricate and demanding notion of sovereignty. If the EU is not to just impose and dominate, there is a need to continuously reflect on and make transparent the grounds on which exercising power over those within the scope of its claims to digital sovereignty is legitimate.

A second point concerns the extent to which being the result of digital sovereignty can make a technology trustworthy and human-centric. The EU seems to assume that there is a set of values specific to it and shared by all EU citizens. Digital sovereignty would then have the goal of preserving or protecting these values. The problem is that it cannot be assumed that everyone endorses the same values and agrees on certain moral principles. Of course, the EU can claim to protect human dignity and human rights – in fact, that step is

enormously important. However, the bundle of values and principles that can be derived from such fundamental norms is not simply static and always existing but is itself the result of negotiations. In this sense, making AI ethical and understanding this as an expression of digital sovereignty suggests that there is clarity about a communal canon of values, and this suggestion will hardly stand up to scrutiny. The philosopher Jacques Derrida pointed out that the question of a possible *centre* of Europe is difficult to grasp (Derrida 1992): Is the *centre* of Europe a city? If so, is it Rome, Paris, Brussels? Is it a language – Latin, French, or English? Or is it a certain common historical narrative or value orientation? These, too, have been many and varied in the history of Europe: the unity of Christianity, the colonisation of other countries and regions, democratisation, respect for human dignity and human rights. In other words, the idea of a firm foundation of moral agreements carries an ambivalence, a tension that always threatens to make the set of values implicated in European digital sovereignty appear to be a fiction, a utopia. This is a feature, not a problem, but it constrains what we can claim in the name of a distinctively European sovereignty.

A third point concerns the procedural level of digital sovereignty: the extent to which the governance rules proposed by the EU can be implemented at all. To achieve this, the EU itself is dependent on the member states ratifying and implementing the proposed governance regulations. The geopolitical claim to power that the EU is trying to make here by means of a certain recourse to values and legal principles thus contains elements of ambiguity. It is not only the procedural preconditions that precede successful implementation within the EU; whether the moral values mentioned by the EU will be recognised as equally central by all EU member states also remains an open question. Doubts here at least do not seem completely far-fetched, as is clear from the debates on the significance of the rule of law principle, among others. In a union such as the EU, in which there is overlap among values but also a significant degree of pluralism of values, constant negotiations about implementation will always be necessary.

This is exactly where the opportunity could lie to frame, shape, and ultimately ameliorate our notion of European digital sovereignty. On the one hand, the goal is to promote European digital sovereignty, which includes the shaping of technology governance on the basis of European values. On the other hand, we observe some overlap in values among European countries but also significant degrees of divergence, contention, and ambiguity. To channel such value pluralism into policy that *both* is an expression of European digital sovereignty *and* results in concrete and operationalisable policy, it is key to uphold certain procedural conditions that guide the process of spelling out values and principles and shaping governance and regulation on this basis. With such conditions in place, the upcoming negotiation and ratification processes as well as the ongoing debates about which values could serve as a basis for a common social fabric can be an expression of the exercise of digital sovereignty.

Consequences for the Governance of AI

Understanding European sovereignty from the point of view of its dependence on procedural conditions also opens up new perspectives with a view to currently emerging debates on the governance framework.

The first point is the enabling of concrete and broad participation, which would ensure that all stakeholders have their views represented and considered. The process of developing the governance framework so far shows that the EU certainly desired and made possible the early integration of different and quite controversial feedback. However, this process cannot be over with the publication of the first draft but rather should be intensified. It could be an expression of European digital sovereignty to anchor specific opportunities for civil society participation in the governance framework itself. This could mean establishing concrete evaluation processes for the effectiveness of the corresponding measures and tying them to specific forms of participation (Stix and Maas 2021). At the same time, it will be essential to enable alternative forms of participation to an even greater extent than before. If it is true that AI will lead to profound transformations of societies and social spaces, the fact that this will also be accompanied by profound transformations of economic and social power must be taken much more seriously than before (Birhane 2021). To shape these transformations responsibly and in alignment with European values, allowing a more direct say to those groups that have been marginalised or underrepresented in existing debates and disputes will be central.

As a second point, transparency in the deliberation process would ensure that all affected parties understand the rationales and reasonings behind decisions and could dispute and contest them based on such an understanding. This relates to two aspects of the governance framework: certain applications of the technologies are each assigned a risk. In many places, however, there are *exceptions*, such as the fact that certain forms of massive surveillance are to be permitted for the purposes of terror surveillance. Exceptions that are insufficiently justified and/or the result of pseudo-deliberative processes threaten to be arbitrary and undercut the reasonableness of the claim to sovereign power that accompanies such policy. Justifications of these exceptions and, at the same time, transparency about how these exceptions came about have so far been very limited. To ensure the trustworthiness of the governance framework, it is crucial to understand why exceptions to rules are to be accepted and on what grounds as well as who will determine them. Will there be more exceptions in the future, or will existing ones be withdrawn? At these points, one may pursue different political agendas.

However, for a Europe that is aware of its relational dependence for the execution of its digital sovereignty, it will be central to make the definition of exceptions and rules as transparent as possible.

Digitally Competent – Digitally Sovereign – Digitally Civic: geopolitics of Subject Formation in the German Context

Jan Winkler, Finn Dammann

Introduction

Since the early 2010s, discursive signifiers such as "digital" or "technological sovereignty" have been circulating widely within political debates in Germany. "Digital sovereignty" is articulated as an answer to the loss of national IT security and global competitiveness of the domestic digital market and to the perceived dependence of state and society on foreign big-tech corporations. Apologists for "digital sovereignty" are thus calling for more state control over the processes of digitalisation and for the economic empowerment of domestic digital companies (see the intro to this forum, Glasze and Dammann 2021; Pohle and Thiel 2020). As studies concerning different geographical contexts have shown – e.g. on Russia (see Pétiniaud, et al. in this forum; Nocetti 2015), China (Budnitsky and Jia 2018) and on the EU (see Glasze et al. in this forum; Floridi 2020) -, problematisations of a 'sovereignty in the digital' are often linked to (geo-)strategic approaches to a regulation of digitalisation processes and data circulation, partly through surveillance and partly by setting technical standards.

In Germany, apologists for digital sovereignty often push for sociotechnical and legal regulations of data circulation (Glasze and Dammann 2021). In addition, another form of "governing" (Foucault 1982) the digital transformation is emerging in Germany that operates by facilitating specific behaviours and capacities of "citizens". These efforts manifest as a "conduct" (*ibid.*) of individuals' economic, social, and political self-conduct within digital fields of action. The highlighted objective of the digital sovereignty of "German citizens" is then linked to questions of what capacities individuals would need to contribute to policy goals such as national IT security, economic competitiveness, and social cohesion.

We argue for analysing the contours of that new figure of a *"digitally sovereign subject"*. Which competencies but also (political) dispositions are linked to this subject – one who is seemingly expected to become digitally sovereign in technical, economic and state-political terms simultaneously (e.g., in terms of enacting civic engagement with regard to perceived challenges of digitalisation)?

Through which sociopolitical technologies a subjectification of individuals as "digitally sovereign citizens" is mediated and how this subject production is embedded in geopolitical power configurations are questions still to be adressed.

Our contribution highlights the relevance of these questions. It is primarly based on an analysis of economic, consumer and educational policy papers on "digital sovereignty" in the German context published in recent years by state institutions, business associations and expert bodies.

Governing Digital Transformation through Population and Subjects

The increasing circulation of data and digital technologies since the 1990s has been accompanied by debates about the question of what skills individuals need in order to deal with digital media and technologies safely and productively. Various speakers in Germany were and are calling for educational measures to foster digital media and technology competencies within the population. Such strategies have been set partly in opposition and partly in addition to technical forms of digital regulating (Glasze and Dammann 2021). Since the early 2000s, calls for more media competencies and for responsible self-regulation in the production, use and circulation of digital content have emerged in political and media debates (Staatsminister für Kultur und Medien 2010).

Attempts to govern digital domains of communication and action by shaping individuals' capacities to think and act – and thus by subjectifying technologies (Foucault 1982) – have also become important in recent problematisations of "digital sovereignty" in Germany since the 2010s. What is problematised are characteristics that "digitally sovereign" individuals need in order to deal with digitalisation competently, autonomously and productively and simultaneously as "responsible citizens" (BITKOM 2019; BMBF 2019; FZI 2017; SVRV 2017). The counterimagination here is a notion of subjects who have no confidence in digitalisation and whose capacities for self-determined and reflective navigation within digitally mediated spaces are diminished – due, e.g., to a perceived loss of control, inadequate technical skills and understanding, or being too heavily influenced (as consumers/users/citizens) by foreign digital economy "big players" (FZI 2017; IIT & Wittpahl 2017; SVRV 2017). The frequent emphasis in the analysed policy papers on concepts such as (personal) self-determination, self-control, freedom, and security implicitly suggests a crisis of the autonomous subject, which is articulated as what is "at stake" in the discourse on digital sovereignty (cf. BMBF 2019; SVRV 2017). In a strategy paper published by the Bavarian Business Association (VBW) that explores possibilities of digital education, this imaginary goes so far as to problematise digital transformation as a kind of dissolution of the "familiar", humanistically enlightened civic subject, a subject identical to itself as an autonomous centre of self-reflexive action. In terms of educational policy, these challenges are countered by strengthening a certain

"personality type" (VBW 2018, 9), "which is characterized by digital sovereignty" (*ibid.*) – an "image of the human being [. . .] who productively absorbs and shapes digitalisation, but also critically processes it" (*ibid.*). This educational focus on the digital literacy of the population and on personality development echoes older debates on media literacy (SVRV 2017; VBW 2018).

In the discourse on digital sovereignty, individuals are located at the intersection of heterogeneous subject positions. They are encouraged to act securely and self-determinedly as consumers, as "consumer citizens" (SVRV 2017, 4), as coproducers of digital content, and as digitally sensitised citizens. Accordingly, in addition to technical, communicative, and economic competencies, political and social dispositions are emphasised: by referring to certain values, individuals should see themselves as civic members of a national community, be aware of their duties and rights, and be empowered to adopt a responsible attitude towards digitalisation. Therefore, digitally sovereign subjects should also be civic, with the ability to "freely, competently, and responsibly use digital media and services [and] actively participate as a citizen in a digital society" (SVRV 2017, 4; BMBF 2019).

These ideas of a "digitally sovereign" subject who first, deals productively with digital technologies and second, sees him-/herself as a citizen of a German or European (value) community (on references to values in EU digital policies, see Braun & Hummel in this forum) are related to both economic and security policy goals. The promotion of a digital economy is thought to "require, in particular, sovereign citizens as co-producers of digital value chains" (BITKOM 2019, 10). The German Competence Center Public IT (KZIT) defined digital sovereignty as "the sum of all abilities and possibilities of individuals and institutions to be able to exercise their role(s) in the digital world independently, self-determinedly and securely" (2017: 3). In terms of security, significance is given, e.g., to the technical knowledge and awareness of individuals regarding browser settings, security updates or data-saving alternative applications and to a critical consciousness of data extractivism and commercial or surveillance strategies of digital economic or state actors (processes that could undermine self-determination) (BMBF 2019; IIT & Wittpahl 2017; KZIT 2017; SVRV 2017; VBW 2018). Essentially, all citizens are called upon to participate in national security in the mode of "self-responsible action" (KZIT 2017, 18).

In addition to security, however, and tied to notions of a value-based political community, a variety of sociopolitical programmes have been articulated in Germany since the late 2010s in reference to "digital sovereignty". Alongside digital education programmes in schools, universities, companies and administrations are further vocational training, technology education and the promotion of civil society self-help activities, such as computer

mentorships or digital "self-help groups" in which older individuals learn how to use digital technologies in order to prepare for their role as digitally sovereign citizens (Stubbe, Schaat, and Ehrenberg-Silies 2019, 44).

Outlook: appeals to a Digital Nation?

While studies on "digital sovereignty" to date have focused on legal and technical-infrastructural mechanisms for regulating data flows, our contribution highlights another, thus far little-explored field of governing: the production of "digitally sovereign" subjects. Attempts to govern digitalisation by fostering autonomous subjects with specific dispositions raise a number of questions. Given the diagnosis that digitalisation affects all aspects of life (BMBF 2019), the desired "digitally sovereign subject" is confronted with numerous expectations (technical and entrepreneurial knowledge, security awareness, responsibility, self-reflexiveness). Moreover, calls for "digital sovereignty" mobilise notions of the "German" or "German-European" nation as a reference and legitimisation of strategies to govern digitalisation. Individuals are supposed to contribute, via willingness to learn, self-control, and responsibility, to the competitiveness of the domestic digital market, to national security, and ideally also to ensuring social cohesion (by self-responsibly tackling digitalisation-related social challenges). Alongside the question of in what heterogeneous fields of action and through what technologies the interpellation of digitally sovereign subjects takes place, the power effects (of inclusion/exclusion), which arise when digital sovereignty is anchored at the level of subjects, must also be taken into account. How do individuals deal with these heterogeneous expectations? Is the new irresponsible "anti-citizen" (Rose 2000) "digitally unsovereign" (whether due to lack of competence and commitment or a political attitude)? Who defines in which contexts digital sovereignty is a goal and how to reach it? Furthermore, the extent to which "digital sovereignty" is made a desired identity of the nation, the community, or the subject, and how it can potentially be effective both as an instrument of power and as an (locally enacted) empowerment strategy, must be examined (Lynch 2020).

The entanglements of digital sovereignty and the notion of critically self-reflective subjects that we observed in the policy documents are also geopolitically configured and intertwined with nation-building processes. In a situation in which non-European digital "big players" are problematised as being too influential, digitally sovereign subjects in Germany and Europe are to develop a critical awareness of these influences (for example, data extractivism) and to challenge those "big players" by asserting civil autonomy. The mobilisation of concepts such as civil autonomy stemming from a liberal discursive archive is linked to new imaginings of a digital economy based on "European values" and positioned as a geopolitical alternative to, e.g., U.S. or

Chinese models (Pohle 2020). Here, how references to (what) values shape the problematisations of the attitudes and behaviours of subjects in the context of the desired "digital sovereignty" remains to be investigated. Despite an assumed geopolitical and economic narrowing of value-related articulations (subjects should be critically-reflective in a way but certainly should not criticise data capitalism), the question of the extent to which recourses to values (a supposedly more liberal or democratic digital economy) could open up discursive conjunctions for more fundamental critical reflections remains. For example, the discourse on "digital sovereignty" seems to allow for mobilisations of issues of social cohesion. Just recently, an alliance of civil and social organisations formed in Germany, calling for a digitally sovereign *society* and stating that "digitalisation is not a purely technical issue, but first and foremost a social issue: it is about [. . .] justice, solidarity, participation and access".[7] Although how socially progressive this project will be cannot be assessed at this point, it at least embraces the digital sovereignty discourse beyond the technical or economic aspects (IIT & Wittpahl 2017; Stubbe, Schaat, and Ehrenberg-Silies 2019). However, decidedly emancipative, participation-oriented local political projects also relate to the signifiers of digital or technological sovereignty, "promot[ing] forms of collective reflection and experimentation with digital systems that challenge de-politicized notions of good digital citizenship" (Lynch 2020, 2; 2019). This demonstrates the potential openness of this discourse and calls for future research on the multiplicity of articulations of "digitally sovereign subjects".

Law Enforcement and Access to Transborder Evidence: the Quest for the Exercise of Digital Sovereignty?

Aude Géry, Florian Nicolai

Digital evidence has become indispensable for criminal procedures, not only in facing cybercrime but also in fighting other kinds of crime. The reason is that law enforcement can gain increasingly more information from digital devices such as smartphones, smartwatches and computers and from connected devices such as smart-home and smart-car systems. To gain access to the data of interest, law enforcement authorities often face the problem of the data being stored in other countries. Data have characteristics that make them unique. According to Daskal, they are characterised by their mobility, divisibility, disconnection between the place where they are stored and the one from which they are accessed and the role of third parties in their management (2015). With the development of cloud computing services, users often do not know where their data are located. Additionally, law enforcement agencies, when investigating, might not know the exact location of data (Council of

Europe 2016; Trüg and Mansdörfer 2014). This results in a "loss of location" in the sense that the localisation of data, defined through spatial criteria, might no longer be relevant, thus challenging the exercise of criminal sovereignty.

The Limits of Data Territoriality for Law Enforcement Agencies

Criminal sovereignty, that is, the ability of a state to exercise its jurisdiction in criminal matters, is profoundly enshrined in territoriality. In international law, the rules for territoriality versus extraterritoriality have been set forth by the Permanent Court of International Justice in the *Lotus* case (Permanent Court of International Justice 1927). The court distinguished between normative jurisdiction and enforcement jurisdiction (Stern 1986). It then ruled that normative jurisdiction can be extraterritorial unless prohibited by international law, whereas enforcement jurisdiction is strictly territorial unless authorised by international law. Exterritoriality is thus not unlawful *per se*. It depends on the type of jurisdiction being exercised and on the existence of specific rules of international law.

In regard to collecting evidence, such an act undoubtedly qualifies as the enforcement of jurisdiction. It is thus unlawful to do so in an extraterritorial manner unless it is authorised by international law. Regarding digital evidence, the same applies to the collection of data. Indeed, even if data can be easily moved from one place to another and even if data localisation might be seen as a constraint for law enforcement agencies, data are stored in infrastructures that are located on the territory of states, and these infrastructures are subject to the sovereignty of the territorially competent state. As such, accessing data located on another state's territory is unlawful unless it is authorised by international law. This situation was at the heart of the famous Microsoft vs. Department of Justice case that ended in 2018 before the U.S. Supreme Court could hold its hearings (Daskal 2018). Here, the territorialisation of data collided with the repartition of jurisdiction. Given the steadily increasing relevance of digital evidence, this collision is rapidly becoming an everyday issue. To overcome this problem, states have recourse to different strategies.

The Quest for Solutions to Overcome Data Territoriality

First, states have signed mutual legal assistance treaties to facilitate international cooperation in criminal matters. However, implementing mutual legal assistance treaties takes an amount of time that is at odds with the pace of data. Second, some multilateral conventions have tried to overcome this problem by granting direct access to data stored in the territory of other states under narrow conditions (see, for example, article 32 of the Budapest Convention on cybercrime of the Council of Europe) (for details on the requirements of Art.

32, see Trautmann, Schomburg, and Lagodny 2020) or by facilitating cooperation on data that constitute evidence. For example, the Budapest Convention contains several articles aimed at facilitating the seizure of data in a short time. However, again, this approach to data is territorial. Those measures aiming at "quick-freezing" the data for the requesting state are, due to being bound to domestic law and therefore, e.g., to the necessity of a judge's decision, not always capable of tackling the risk of data's ephemeral character to the success of an investigation.

To address the difficulty of data localisation and thus their territorialisation, states have started to use other mechanisms. The United States adopted the *Clarifying Overseas Use of Data Act* (Cloud Act 2017–2018). It requires any service provider subject to U.S. jurisdiction to transmit data controlled by it or its subsidiaries to U.S. authorities. Here, the U.S. government is circumventing the territorialisation of data by using a basis other than territoriality to establish its jurisdiction and collect data overseas. It invokes the nationality of the third parties that host data, and because nationality is a valid criterion for invoking and exercising jurisdiction, the law is consistent with international law. It should, however, be noted that the way that the United States interprets the criterion of nationality is very extensive, making its lawfulness under international law questionable.

The Cloud Act also provides for the possibility of the U.S. government entering into intergovernmental agreements facilitating the collection of data from service providers located in the jurisdiction of the other state party to the agreement. For example, the United Kingdom and the United States entered into an executive agreement on access to electronic data for the purpose of countering serious crime in October 2019. The agreement "*places obligations on the two countries to remove the barriers in domestic law within the jurisdiction the CSP [content service provider] is located which could otherwise prevent disclosure of this data recognised by this Agreement*" (United Kingdom 2019).

On November 17, 2021, the Council of Europe adopted the 2nd additional protocol to the Budapest Convention on electronic evidence. It aims at facilitating access to electronic evidence, notably by providing a legal basis for direct cooperation between LEAs and service providers (article 7).

At the EU level, the European Commission has proposed a regulation to facilitate the collection of electronic evidence in criminal cases. This so-called *E-Evidence* draft plans to give law enforcement the right to directly request data from service providers if the data might be of interest in criminal cases. This draft has been controversial (see, e.g., Burchard 2019) since it originally designated service providers as those responsible for providing access to the requested data without the involvement of local law enforcement (European Commission 2018). In her draft report, EU Parliament Rapporteur Birgit Sippel 2018 proposed a wide range of amendments to the draft, *inter alia* calling for better notification systems, better rights for the affected states and

better protection of human rights (2018). The draft is currently being discussed in the trilogue between the European Parliament, Council and Commission, but this demonstrates that territoriality is still an important criterion.

States are currently seeking other legal underpinnings on which to base their jurisdiction or internationally agree to allow direct access to evidence. This reveals a decline in the territorial criterion of data, which is increasingly deemed irrelevant by many states. This is true even outside the realm of cybercriminal matters. For example, the G7 Communiqué (2021) released on June 13, 2021, endorsed the *G7 Roadmap for Cooperation on Data Free Flow with Trust* adopted by the G7 digital and technology ministers that subtly chose free data flows over data localisation laws (G7 Digital and Technology Track 2021). The *Roadmap* questions data localisation laws, that is, national legislation that requires companies to store data of a state's citizens within its borders. from different perspectives. Regarding government access to data, the *Roadmap* states that states "*are committed to maintaining domestic data protection and privacy standards, reasonable principles underpinning lawful access regionals, as well as legal powers and arrangements that facilitate access across borders*". The phrasing of the sentence expresses the search for a balance between the territoriality of citizens and the government's need to access data. It goes further by stating that any access must be allowed on "reasonable principles". It could be read as a covert criticism of the large interpretation of the criterion of nationality by the United States in the context of the *Cloud Act* or as a reminder that human rights must be taken into account. The *Roadmap* thus tries to circumvent data localisation both for economic reasons and to preserve states' right to enforce their criminal jurisdiction.

A Trend that Is Not Shared across the World

However, this trend is not shared by every country around the world. For example, some states have, over the past few years, adopted data localisation laws. These laws can be applicable to "important data", such as the Chinese law on data security (Sacks, Chen, and Webster 2020), or to data belonging to a state's citizens, such as the Russian personal data localisation law (Savelyev 2016). India has a rather ambiguous position on the criterion of data territoriality. On the one hand, it has adopted data localisation regulations for specific sectors (Burman and Sharma 2021). On the other hand, it has recommended a "*new form of sovereignty which should be based on ownership of data" instead of where the data is physically located* (India's comments on the Initial Pre-Draft of the report of the OEWG on developments in the field of information and telecommunications in the context of international security 2020). This seems to refer not to the territorial criterion but to the criterion of the nationality of India's citizens. This example illustrates the dilemma that states might face

when apprehending the territoriality of data and the exercise of their sovereignty: adopting the traditional view that jurisdiction is first and foremost defined by a state's territory while needing to face the fact that data can be located throughout the world. It is even truer in the context of the collection of evidence by law enforcement agencies.

Conclusion

One of the visible ways for a state to exercise its sovereignty is its ability to pursue criminal offences over which it has jurisdiction. The repartition of data throughout the world challenges the conduct of criminal investigations and thus the right of a state to enforce its criminal jurisdiction. In this sense, data challenge state sovereignty. To preserve their ability to exercise their sovereignty in criminal matters, states have been negotiating international agreements to facilitate access to evidence located outside their territory and have sought other legal underpinnings on which to base their jurisdiction. This trend is not shared by every country around the world, as some states still attach great importance to the localisation of data. Furthermore, the observed trend should not lead to the conclusion that states renounce the territorial criterion in the digital space or to sovereignty. These things still matter, especially for the protection of the human rights of people whose data are being collected. Here, the European states occupy a special position. They have adopted (or are in the process of adopting) all the tools and practices mentioned above while keeping a territorial approach to protect their citizens' rights. In addition, they are doing so in a way that aligns with the exercise of sovereignty because, *in fine*, negotiating agreements to facilitate the collection of data is an act of sovereignty.

Platforms Vs. States: A Sovereignty Conundrum

Stéphane Grumbach, Caroline Zanin

The Revolution of Mediation

Intermediation Platforms

In the last two decades, a new category of actors has emerged: digital platforms (Constantinides, Henfridsson, and Parker 2018; Gawer 2014). They have experienced a unique growth in economic history, becoming the world's largest corporations in terms of both number of users and market capitalisation. Their emergence constitutes a radical socioeconomic

transformation (Choudary, Parker, and Van Alstyne 2015) of the activity of mediation (Frenot and Grumbach 2014; Van Dijck, Poell, and de Waal 2018).

Intermediation is an essential function of any society and is ubiquitous in satisfying most of our needs: consumption, health, education, etc. The role of intermediaries is basically to connect people in two-sided markets (Rochet and Tirole 2003) with the goods and services they need, which are provided by producers. Health organisations, for instance, rely on a complex nexus of patients and the countless actors offering care, drugs, research, insurance, etc. What changes with digitalisation is the appropriation of the mediation function by digital players that ensure not the service but only the connection and are remote from the physical world.

In these processes, information plays a key role, allowing intermediaries to identify needs and provide the most relevant services. Digital platforms handle information through comprehensive and continuous interactions with users. By harvesting and analysing data produced mostly outside the system, platforms are able to develop essential mediation services. The use of these services generates new data and traces of users' activity, which enable platforms to generate additional services, including generic and personalised services, which produce even more data, thus enabling platforms to exploit positive feedback loops to ensure their growth (Figure 1).

Intermediated Dimensions

Through such positive feedback loops, platforms penetrate society even more deeply in an unbounded number of sectors as diverse as commerce (Amazon, Aliexpress), education (Microsoft Teams), the press (Google, Facebook), health (Apple Watch, Exposure Notification), and security.

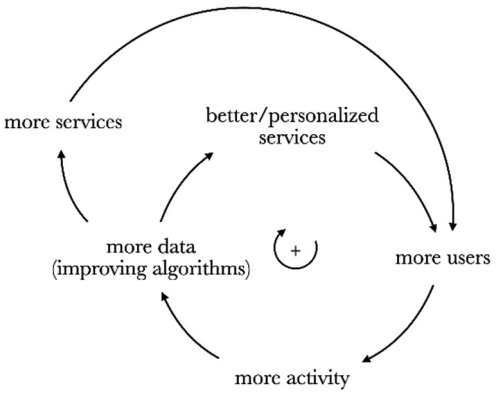


Figure 1. Feedback loop model.

The growth of Uber is emblematic of such phenomena and their socio-economic consequences. Through multiple feedback loops (Gurley 2014), Uber offers a system that is more efficient than that of traditional taxi companies, both for the consumer and the driver, and, as a side effect, "disintermediates" the consumer and driver (Hawkins, Mansell, and Steinmueller 1999). As Uber has grown, it has developed additional services, such as Uber Rush (courier service), Uber Eats (food delivery), and Uber Essentials (online ordering and delivery) as well as free vaccine rides (Uber Newsroom 2021a) to facilitate Covid vaccination policies, thus disintermediating legacy actors in an increasing number of dimensions. In doing so, it also affects social norms and standards.

In a sense, platforms draw their strength from the very limits of the economic model they are attacking, and this transformation is not uncontroversial. This point is all the more relevant as platforms gradually disintermediate public administrations. Through comprehensive, continuous real-time and convenient interactions with people, digital platforms provide intermediation services that public administrations cannot offer and that become essential if not necessary for the functioning of society as a whole. The fact that Uber has offered a vaccine reservation service as well as a ride to fight against the Covid-19 pandemic (Uber Newsroom 2021b) is just one example among many (Google Flu Trends, Exposure Notification). Thus, from initially insignificant services (lift services), digital platforms end up managing fundamental services and setting the new norms on which society relies.

The Globalisation of Flows in the Technosphere

The intermediation platforms used by the population are but the visible part of a system controlling the flows of matter, energy and information at the global scale.

The geologist Peter Haff (2014) first proposed considering all human activities as a holistic system: mining resources, energy, transforming industries, agriculture, transportation, etc., together with the rules (laws, norms, tariffs, monopolies, embargos, etc.) that bind them. The "technosphere" forms a complex system of interdependent actors in which algorithmic intermediation constitutes the beating heart.

This system results in the appropriation of natural resources and global flows of matter, controlled by an extremely distributed system but in which very powerful intermediation actors control exchanges in fundamental sectors, such as the financial system, raw material market places, maritime transportation, and energy. The financial transaction platform SWIFT provides a good illustration of how an actor can acquire an extremely central position. Until the 1970s, cross-border transactions between banks relied on public telegram and telex systems, which were operated by national telecommunication providers. These systems proved both slow and insecure. Such inefficiencies led financial actors

to create a platform for interbank communication in order to build a messaging system that could replace public providers and speed the payment process. SWIFT then became dominant for international transactions.

Like for the growth of Uber, SWIFT benefited from the network effect. The more banks used SWIFT, the more it created measurable network benefits for its members (Scott, Van Reenen, and Zachariadis 2017). Even more financial institutions began to use and become dependent on the SWIFT system. As of 2015, the system linked more than 11,000 financial institutions in more than 200 countries and territories, which exchange a million messages per day. The emergence of such a monopolistic position has had important geopolitical consequences because it can be constrained to obey injunctions from governments, either to access information or to block services for certain actors. The initial secrecy of SWIFT activities was reduced in the fight against terrorism, and coercion was exerted by the United States, which excluded Iran from the system in 2012.

Evolution of Global Imbalances and Asymmetries

Typology of Sovereignty

As a consequence of their growing importance in an increasing number of social dimensions, the emergence of digital platforms has also induced a political revolution. Such platforms now have a global reach; they bring together millions, even billions, of users around new norms and new actions. They have led to the emergence of new equilibria.

The structure of cyberspace, described as a "complex arena" by Joseph Nye (2014), relies on a complex system of interdependent and overlapping actors and institutions. Like many other complex systems, cyberspace tends to generate asymmetrical topologies (Allen 2011) in which flows are centralised, passing through a few specific intermediaries (Pradgett and Ansell 1993). In this "hub and spoke" system (Strange 1996), some actors, or "nodes", occupy a central position, while most other nodes depend on them. Once these asymmetries are established, it is difficult for other actors to modify them; on the contrary, feedback loops tend to tighten them.

By establishing powerful centrality in networks, platforms de facto provide states with levers to extend their power and influence beyond their borders (Nye 2014). A multidimensional concept of sovereignty thus arises. It still has physical and territorial dimensions (e.g., physical data storage) but also acquires a technological dimension (e.g., control of technical infrastructures) and a legal dimension through the potentially extraterritorial legal scope of norms (e.g., Cloud Act, Patriot Act) or normative influence of digital actors (e.g., Uber and work laws).

It is also a question of influence of soft power (Nye 2011) (e.g., news and truth construction in a territory). All these dimensions must be considered in the strategic projections of states into renewed forms of sovereignty in the digital era.

The coexistence of these different aspects of sovereignty leads to conflicts between orthogonal orders and raises fundamental questions: Who sets the rules in cyberspace? Who decides to censor something or someone or to exclude a nation from a global service? Who determines what information should be shared on social networks? Who keeps and uses users' personal data? On what legal basis and with what legitimacy?

State and Platform Relations

Such a shift of power results in increased interdependency among actors. Platforms are weakening states in their regalian domain. They provide a number of essential services that would have been offered under the initiative or at least supervision of governments in the past, while states have become both dependent on and in competition with them.

Equilibria are established in each territory between the central government, digital platforms and regulation (legislators, activists), raising an essential question of power and competence. The negotiations between platforms and the U.S. government to establish a collaborative policy on national security issues allowing the use of user data illustrates these new balances of roles and power.

However, the consequences of platform dependency may become even more critical at a global scale. As shown by Farrell and Newman (2019), platforms can become surveillance assets and asymmetric coercive tools. One of the most striking examples, according to these authors, is the "weaponization of SWIFT" by the U.S. government during the Iranian crisis, considered above. Because of the centrality of the United States in the international payment system, SWIFT offers the U.S. government a significant coercive power: the capacity to disconnect. In 2012, the Iranian regime and its major financial institutions, including its central bank, found themselves locked out of the international payment system. In February 2022, President Biden signed an executive order to use the billions of assets from Afghanistan's central bank frozen in the US. This capacity to unplug can be exerted at all granularity levels, from nations to people. Legitimately elected politicians in several countries have experienced this sudden exclusion from the social networks decided by platforms.

Typology of Nations and Digital Empowerment or Disempowerment

Platforms necessarily rely on territories, as their headquarters are connected to a given state; therefore, some geopolitical asymmetries develop in cyberspace in terms of sovereignty and strategic power balances that result from the nonuniform distribution of platforms across the world. We propose to distinguish three types of states.

Full sovereign states, such as the United States and China, are essentially independent and autonomous in terms of their digital infrastructures, technologies and contents. Owing to their domestic institutions and norms, e.g., the U.S. Patriot Act (2001) and Cloud Act (2018) and China's cybersecurity law (2017), they can deploy their power and influence in numerous dimensions, including norms and standards, physical infrastructures, technologies, global surveillance and intelligence, regulations, and ideological influence, thus shaping a coercive authority beyond their borders.

Balanced states, such as Korea and Russia, are partially independent and autonomous in terms of their digital infrastructures, technologies and contents and project their influence on a mostly regional scale. Russia has an almost complete ecosystem of platforms and services founded by Russians and governed by Russian law. As shown by the contribution of Pétiniaud, Limonier and Bertrand, the RuNet services provide it with a dominant position in the post-Soviet digital networks (see also Limonier 2018).

Finally, dependent states, such as European nations, are almost completely dependent on foreign platforms, technologies and infrastructures and enjoy no digital sovereignty. The introduction of the General Data Protection Regulation (GDPR) in 2018 demonstrates the uniqueness of the European position, which is guided more by a legal than a political approach to the digital realm. The fact that a large majority of European countries chose to use Apple's and Google's exposure notifications technology to develop contact tracing applications during the Covid-19 pandemic is representative of this status of "digital colony" (Morin-Desailly 2013).

Conclusion

The growing power of digital platforms has increased the intricacy of the dependencies among actors and contributed to the emergence of new modes of governance and new global powers and the disruption of established ones.

This revolution raises very profound political questions regarding the legitimacy of the forms of government and the balance of power among actors. However, it is occurring at a time when sovereignty is already at stake for a much more fundamental reason: the perspective of environmental degradation and its consequences for geopolitical equilibria.

Human society is becoming more complex, it must progressively integrate environmental parameters to measure all its activities, adapt to less favourable global conditions and mitigate further degradation (Grumbach et al. 2018). This implies monitoring the technosphere in radically new ways, from ecosystemic dimensions to individuals' private sphere. Intermediation platforms

are clearly well positioned in this race, which is social, economic and political, to guide or force humanity to a new world order that integrates the global constraints imposed by planetary boundaries.

Notes

1. In the parallel area of education in engineering and computer science for instance, Russia can also be considered as an important hub, especially for Central Asia.
2. The Baltic countries, most notably Estonia, have developed their Internet specifically avoiding Russian influence.
3. Since Benhamou and Sorbier's article, the role of the US in the Internet Governance process has evolved. Under the Obama administration, the US agreed to step back of their historical role in the ICANN, one of the principal organisations in the Internet Governance process.
4. President Macron has proclaimed "strategic autonomy" to be a leitmotif of his European policy – see for example his extensive interview on this subject in 2020 (https://legrandcontinent.eu/fr/2020/11/16/macron/ – English translation see: https://geopolitique.eu/en/2020/11/16/the-macron-doctrine/; 23.12.2021).
5. Among many other similar voices, the French foreign minister in a major speech in 2019 (https://www.vie-publique.fr/discours/272268-jean-yves-le-drian-06122019-union-europeenne; 10.01.2022) and the German chancellor in her government declaration in 2020 call for a digitally sovereign Europe (https://www.bundeskanzler.de/bk-de/aktuelles/regierungserklaerung-von-bundeskanzlerin-merkel-1762594; 10.01.2022). In this context, it is often implicitly assumed that (German or French) national and European interests coincide, which, given the dominance of voices from France and Germany in this debate, would at least need to be explored more closely.
6. See for example the State of the Union address of Commission President Ursula von der Leyen in 2021 (https://ec.europa.eu/info/sites/default/files/soteu_2021_address_en_0.pdf; 23.12.2021) and the integration of cyber-defence issues in the process of building a strategic compass for the EU https://eeas.europa.eu/headquarters/headquarters-homepage/106337/towards-strategic-compass_en; 10.01.2022).
7. https://digitalezivilgesellschaft.org/ (01.08.2021)

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